Chapter 3

Cyclic Groups

Let **G** be a group with operation $*$ and let **H** be a subset of **G**. We say that **H** is **closed under** $*$ if

$$a, b \in \mathbf{H} \quad \text{implies} \quad a * b \in \mathbf{H}.$$

If **H** is closed under $*$, then **H** also forms a binary system. If **H** turns out to be a group, we call it a **subgroup** of **G**. Notice that if **H** is closed under $*$, then **H** is automatically associative. Thus **H** is a subgroup of **G** if and only if

(i) **H** is closed under $*$,

(ii) the identity e of **G** is in **H** and

(iii) for each element b of **H**, the inverse b^{-1} is in **H**.

For example, the subgroups of the additive group **R** of **real numbers** include the additive group **Q** of **rational numbers** and the additive group **Z** of **integers**. Familiar subgroups of the additive group $\mathbf{F}(\mathbf{R},\mathbf{R})$ include the group of **continuous real valued functions** $\mathbf{C}(\mathbf{R},\mathbf{R})$, the group of **differentiable real valued functions** $\mathbf{D}(\mathbf{R},\mathbf{R})$ and the group of **polynomial functions** $\mathbf{P}(\mathbf{R},\mathbf{R})$:

$$\mathbf{P}(\mathbf{R},\mathbf{R}) \subseteq \mathbf{D}(\mathbf{R},\mathbf{R}) \subseteq \mathbf{C}(\mathbf{R},\mathbf{R}) \subseteq \mathbf{F}(\mathbf{R},\mathbf{R}).$$

Our next lemma gives a simple way to find a multitude of subgroups.

Lemma 20. *Let b be an element of the group* **G**. *Then the set* $\{b^n \mid n \in \mathbf{Z}\}$ [*the set* $\{nb \mid n \in \mathbf{Z}\}$ *in additive notation*] *of all powers* [*multiples*] *of b forms a subgroup of* **G**.

We use the notation

$$\mathbf{sg}(b) := \{b^n \mid n \in \mathbf{Z}\}$$

and call this the **cyclic subgroup of G generated by** b. If $\mathbf{G} = \mathbf{sg}(b)$ for some element $b \in G$, we say that **G** is a **cyclic group** and that b is a **cyclic generator** of **G**. For example, the additive group **Z** of integers is cyclic since every integer is a multiple of 1, that is, $\mathbf{Z} = \mathbf{sg}(1)$.

Lemma 21. *Every cyclic group is commutative.*

Thus, for example, $\mathbf{S}_3$ and $\mathbf{M}_1$ are *not* cyclic groups. Notice that each element of every group generates a cyclic subgroup. We will see that different elements can generate the same cyclic subgroup.

Lemma 22. *If* $\mathbf{H}$ *is a subgroup of* $\mathbf{G}$ *and* $b \in \mathbf{H}$*, then* $\mathbf{sg}(b) \subseteq \mathbf{H}$.

Problem 23. *Find the cyclic subgroup of* $\mathbf{Z}_{12}$ *generated by each of its elements. Show that* $\mathbf{Z}_{12}$ *is itself a cyclic group. Is there a subgroup* $\mathbf{H}$ *of* $\mathbf{Z}_{12}$ *that is not cyclic?*

We have seen a small number of different finite groups. How many different finite groups are there? For example, is there a group with 17 elements? a group with 64,539 elements? Looking closely at the group $\mathbf{Z}_{12}$ suggests some answers. Let n be an arbitrary positive integer and consider the set $\{0,1,2,3,\ldots,n-1\}$ of all possible remainders when we divide by n. Following the example of $\mathbf{Z}_{12}$ we define $+_n$ on this set by

$$a +_n b := \begin{cases} a+b & \text{if} \quad a+b<n; \\ a+b-n & \text{if} \quad a+b \geq n. \end{cases}$$

Clearly 0 is an identity for $+_n$ and $n-b$ is an inverse for b. Moreover, $+_n$ is associative since both $a +_n (b +_n c)$ and $(a +_n b) +_n c$ are the number we get by adding $a+b+c$ and then subtracting n until we get a number less than n. Thus

$$\mathbf{Z}_n := \langle \{0,1,2,\ldots,n-1\}; +_n \rangle$$

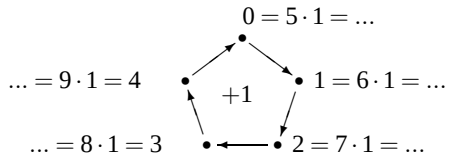
is a (commutative) group which we call the **group of integers modulo** n. This gives us at least one n-element group for *every* positive integer n.

Problem 24. *Show that* $\mathbf{Z}_5$ *is cyclic. Exactly which elements of* $\mathbf{Z}_5$ *are cyclic generators of* $\mathbf{Z}_5$*?*

Problem 25. *Show that the group* $\mathbf{Z}_n$ *is cyclic for each* $n \in \mathbb{Z}^+$*. Looking at the examples of* $\mathbf{Z}_5$ *and* $\mathbf{Z}_{12}$*, which elements of* $\mathbf{Z}_n$ *do you think are the cyclic generators of* $\mathbf{Z}_n$*? State your answer as a conjecture. Can you prove it?*

The cyclic group $\mathbf{Z}_5$ can be illustrated by the following diagram.

$\mathbf{Z}_5$:



This diagram certainly does support our choice of the term "cyclic" for this kind of group. We would like to show that every finite cyclic group looks like this. If a cyclic group has generator b, then it consists of all the powers

$$\ldots b^{-3}, b^{-2}, b^{-1}, b^0 = e, b, b^2, b^3, b^4, \ldots$$

of the element b. If the group happens to be finite, then this list must be highly redundant as it is in $\mathbf{Z}_5$. We will see that this redundancy must follow the above "cyclic" pattern.

Lemma 26. *Let* b *be an element of a group* $\mathbf{G}$*, and let* $n \in \mathbb{Z}^+$*. Assume that the powers* $e = b^0, b = b^1, b^2, b^3, b^4, \ldots, b^n$ *are not all distinct, that is, there are integers* i, j *such that* $b^i = b^j$ *where* $0 \leq i < j \leq n$*. Then there is a positive integer* $k \leq n$ *such that* $b^k = e$*.*

We say that b has **infinite order** if all of the non-negative powers $e = b^0, b = b^1, b^2, b^3, b^4, \ldots$ are distinct. Otherwise, we say that b has **finite order** and we define the **order** of b, written $\circ(b)$, to be the smallest positive integer k such that $b^k = e$ [additively, $kb = 0$]. For example, each element of the additive group $\mathbf{Z}$ has infinite order while each element of a finite group must, according to Lemma 26, have finite order.

Problem 27. *Find the order of the following elements of the group* $\mathbf{M}_1$:

$$\text{(i) } A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \quad \text{(ii) } B = \begin{pmatrix} 3 & 5 \\ 4 & 7 \end{pmatrix} \quad \text{(iii) } C = \begin{pmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}$$

Lemma 26 leads to several useful results. Recall that $\mathbf{H} \subseteq \mathbf{G}$ is a subgroup of $\mathbf{G}$ provided that it is closed, contains the identity and contains inverses. These criteria can be simplified for finite groups.

Theorem 28. *Let* $\mathbf{H}$ *be a finite non-empty subset of a group* $\mathbf{G}$. *Then* $\mathbf{H}$ *is a subgroup of* $\mathbf{G}$ *provided only that it is closed.*

Corollary 29. *If* $\mathbf{G}$ *is a finite group and let* $a \in \mathbf{G}$, *then the subgroup generated by a consists of all positive powers of a:*

$$\mathbf{sg}(a) = \{a^k \mid k \in \mathbf{Z}^+\}.$$

We can now prove two important theorems about cyclic groups. We will need the following fundamental fact about the integers, which we will not prove here.

Long Division Lemma *Let* m *be an integer and let* d (*the* **divisor**) *be a positive integer. Then there are unique integers* q (*the* **quotient**) *and* r (*the* **remainder**) *such that*

$$m = qd + r \quad \textit{with} \quad 0 \le r < d.$$

Let $m, d \in \mathbf{Z}$. We say that d **divides** m, written $d|m$, if there is a $q \in \mathbf{Z}$ such that $m = qd$. We generally use Long Division to find out if $d|m$ by dividing m by d and seeing if the remainder is 0. Use this strategy to prove the following lemma.

Lemma 30. *Assume* $b \in \mathbf{G}$ *has finite order, and let* $m, i, j \in \mathbb{Z}$. *Then*

(i) $b^m = e$ *if and only if* $\circ(b)|m$, *and*

(ii) $b^i = b^j$ *if and only if* $\circ(b)|i-j$.

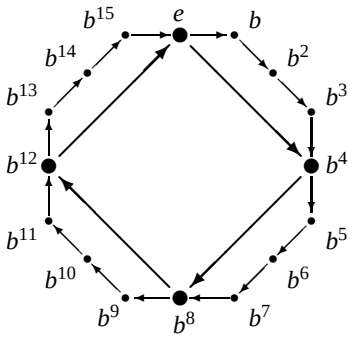
Theorem 31. *Let b be an element of a group* $\mathbf{G}$, *and assume that b has finite order k. Then the cyclic subgroup of* $\mathbf{G}$ *generated by b has exactly k distinct elements, namely,*

$$\mathbf{sg}(b) = \{e, b, b^2, b^3, \ldots, b^{k-1}\} \textit{ with } b^k = e.$$

This theorem tells us that a finite cyclic group is accurately described by the adjective "cyclic". Using this information, we can fill in the entire group table for $\mathbf{sg}(b)$. Thus every group element of order k generates a cyclic group that look exactly like the cyclic group $\mathbf{Z}_k$. The next theorem tells us that what we discovered about $\mathbf{Z}_{12}$ in Problem 23 is true of every cyclic group.

Theorem 32. *Every subgroup of a cyclic group is cyclic.*

Proof: (Hint) Let $\mathbf{G} = \mathbf{sg}(b)$ be cyclic, and let $\mathbf{H}$ be a subgroup of $\mathbf{G}$. Choose an integer n for which you believe that $\mathbf{H} = \mathbf{sg}(b^n)$. Then show that $\mathbf{sg}(b^n) \subseteq \mathbf{H}$ and that $\mathbf{H} \subseteq \mathbf{sg}(b^n)$.



The fact that $\mathbf{Z}$ is a cyclic group leads to some useful properties of the integers. For example, we know that, for each non-negative integer n, the multiples of n form a cyclic subgroup of $\mathbf{Z}$. Theorem 32 tells us that these are the *only* subgroups of $\mathbf{Z}$. Let $m,n \in \mathbf{Z}$ where m and n are not both 0. We say that d is the **greatest common divisor** of m and n, written $d = \gcd(m,n)$, if d is the largest positive integer that divides both m and n.

Lemma 33. *Assume $m,n \in \mathbf{Z}$ where m and n are not both 0. Then $d := \gcd(m,n)$ is the smallest positive integer that can be expressed in the form $mx+ny$ where $x,y \in \mathbf{Z}$.*

Proof: (Outline) Show that $\mathbf{H} := \{mx+ny \mid x,y \in \mathbf{Z}\}$ is a subgroup of $\mathbf{Z}$ and apply Theorem 32.

We say that $m,n \in \mathbf{Z}$ are **relatively prime** if $\gcd(m,n) = 1$.

Theorem 34. *Integers $m,n \in \mathbf{Z}$ are relatively prime if and only if there are integers $x,y \in \mathbf{Z}$ such that $mx+ny = 1$.*

We now have the machinery we need to prove the conjecture of Problem 25.

Theorem 35. *Let $\mathbf{G} = \mathbf{sg}(b)$ be a finite cyclic group where $\circ(b) = k$. Then b^m is a cyclic generator of $\mathbf{G}$ if and only if m and k are relatively prime.*